

CMPT 276 - Phase 4 Report

Project Group 14

Joshua Kim, Tim Yeh, Rodrigo Anasco, Sarah Harb

Hungry Doug is a 2D maze game where the player controls Doug, a dog who must navigate through various maps to collect bones while avoiding enemies and punishments. The player's goal is to collect all bones to make a gem appear, signaling the level's completion. The game includes interactions, such as rewards that provide bonus points, punishments that reduce health, and rats that chase Doug.

The Game

The development of the game remained pretty faithful to its original concept, focusing on Doug, a hungry dog navigating a maze to collect food items while avoiding enemies and finding the exit. The core gameplay mechanics—maze navigation, collecting points through food items, avoiding enemies, and reaching the exit—were retained throughout the process. Player controls were implemented using both arrow keys and WASD, ensuring accessibility and user-friendly gameplay. The final game features rats as the sole enemy type, which actively pursue Doug, adding a dynamic challenge. While the original design envisioned additional enemies, such as cats, and multiple maps with varying environments, these elements were omitted due to time and resource constraints. Rewards, including items like apples, meat, and eggs, align with the original concept of incentivizing exploration, while hazards, such as poisonous items, were simplified but still penalize players by deducting points.



Figure 1: Phase 1 initial game design idea

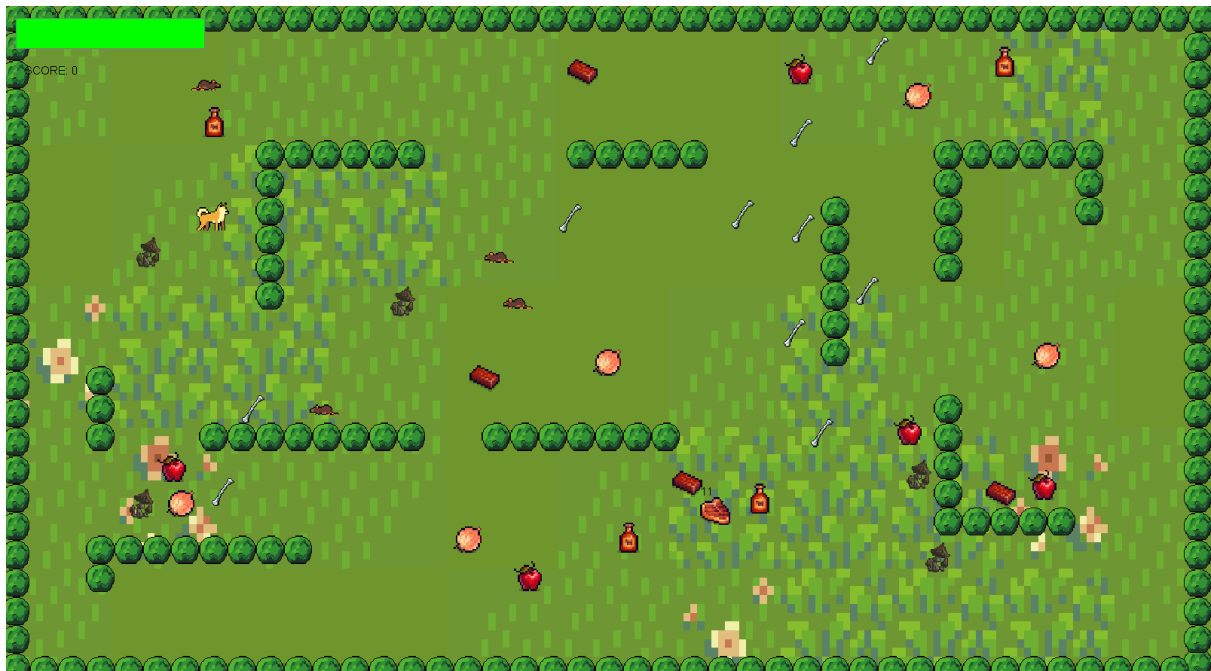


Figure 2: Game design final product

As you can see in the two figures above, we have remained as faithful to the original idea as possible. A few key things to note however. First, unlike the original, we have decided to randomly scatter the various objects across the maze for better replayability of our game, and perhaps adding a bit of a challenge to explore with each run. Secondly, due to difficulties of implementing a well working rat AI, we had to dumb down our maze a little such that the rats can more easily reach Doug wherever on the map they may be. Lastly, some of the sprites we have used differ slightly from the original. This was a collective decision as we decided the current sprites appear a lot better and fitting for our game.

Ultimately, the creation of Hungry Doug was an experience, and each one of us has learned a ton. Initially, we struggled with tools like Git - merging changes, handling large commits, and learning best practices. But eventually it became critical in allowing us to work independently on different aspects of the project while maintaining a cohesive codebase. We often had different ideas on the design and implementation, which required a lot of discussion, talking through solutions, and some compromise. Additionally, we hit a number of dead ends when implementing certain features and had to back track, reroute, or completely start over. If one person got stuck, someone else might try a new approach. The challenges we encountered with this project ultimately provided opportunities for growth and learning. We were able to collaborate better as a team, share ideas openly, and improve our communication.

On the technical side, we became more comfortable with version control tools, Java, and important software design principles. Not only has collaborating on this project deepened our Java skills (such as working with threads and utilizing concurrency, general object and class creation, etc), but leveraged our understanding of object-orientated principles, exception handling and much more. Testing and refactoring were also essential, as we ensure proper functionality of our game, and apply good coding practices. This project was a first regarding Java for many of us, but we were able to adapt, learn, and apply new knowledge. We have all thoroughly enjoyed working on Hungry Doug, and we hope you enjoy our game!

Game Tutorial

See the short video below for example playthroughs of our game, and some instructions as to how to play! (click on the link below)

[VIDEO LINK](#)

Within our menu UI, press *W/Up arrow key* or *S/Down arrow key* to switch between menu buttons, and *ENTER* to click on them. The 'START' button will start the game, 'RULES' give further instructions on how to play, and 'EXIT' will close the game window. Doug will have a total of **200 HEALTH POINTS** with every new game.

Some of the key features of our game are shown below:

   - Bonus rewards, eating them increases your score. Apples and mushrooms are static and appear at the start of the game while steaks appear randomly and disappear on a timer.

-  - Apples provide **10 POINTS**.
-  - Mushrooms provide **10 POINTS**.
-  - Steaks provide **20 POINTS** and will randomly last for **5-25 seconds**.

 - The bone reward; necessary to beat the level, eating all of them unlocks the exit gem. Each bone provides **15 POINTS**.

   - Various punishments appear on the map, eating these will lower Doug health

-  - Chocolates causes Doug to lose **50 HEALTH POINTS**.
-  - Onions causes Doug to lose **20 HEALTH POINTS**.
-  - Whiskey causes Doug to lose **25 HEALTH POINTS**.

 - The rat constantly chases the player, touching it will deplete **ALL YOUR HEALTH POINTS**, resulting in a loss.

 - Bush obstacles stops both players and rats from moving through it.

 - The exit gem appears after all bones spawns have been consumed.